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FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

APPLIED THERMODYNAMICS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	a.	With suitable diagram list the important processes in Carnot cycle.	L1	CO1	**02**
	b.	Differentiate between two stroke and four stroke diesel engine.	L1	CO2	**02**
	c.	Write down the volumetric efficiency equation for an air compressor with suitable nations.	L1	CO3	**02**
	d.	Lit the important factors which affect the nozzle efficiency.	L1	CO4	**02**
	e.	Compare the heat engine and refrigerator with suitable sketch.	L1	CO5	**02**
		PART-B	**6 X 5=30**		
2.	a.	Explain the simple Rankine cycle with neat sketch and PV diagram.	L2	CO1	**05**
		OR			
	b.	An air standard Otto cycle has compression ratio of 8. The temperature and pressure at the beginning of compression are 300 K and 102 kpa. If the maximum cycle temperature is 1900 K. determine heat supplied per kg of air. Assume γ=1.4.	L3	CO1	**05**
3.	a.	With a neat diagram explain the combustion in CI Engine. List effects of knocking.	L2	CO2	**05**
		OR			
	b.	Explain the working principle of thermo-syphon cooling system for an engine.	L2	CO2	**05**
4.	a.	With a neat sketch and P-V diagram explain the working principle of multi-stage reciprocating air compressor.	L2	CO3	**05**
		OR			
	b.	Write the difference between the reciprocating and centrifugal air compressors.	L2	CO3	**05**
5.	a.	Distinguish between impulse and reaction turbine.	L2	CO4	**05**
		OR			
	b.	Air is allowed to expand reversibly and adiabatically in a nozzle from 13 bar and 150 ℃ to a pressure of 5 bar. The inlet velocity of the nozzle is very small and the process occurs under steady state flow conditions. Calculate the exit velocity of steam at nozzle.	L3	CO4	**05**

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6.	a.	With a neat sketch, P-V & T-S diagram explain the working principle of vapor compression refrigeration system.	L2	CO5	**05**
		OR			
	b.	A refrigerator works on Bell-Coleman cycle between the pressure limits of 100 kpa and 400 kpa. Air leaves the refrigerator at 60 °C and the cooler at 32 °C. The compression and expansion follow the law $PV^n=C$. Assume Cp=1.005 KJ/kgK and γ=1.4 for air. Determine COP of the cycle.	L3	CO5	**05**
7.	a.	Explain the important properties of a good refrigerant. List the merits and demerits of refrigerants.	L2	CO5	**05**
		OR			
	b.	Explain any one summer air conditioning systems with neat sketch.	L2	CO5	**05**
		PART-C	**6 X 10=60**		
8.	a.	Derive expression for efficiency of Diesel cycle with a P-V diagram.	L3	CO1	**10**
		OR			
	b.	The temperature and pressure of a Diesel cycle at the end of constant volume heat rejection is 50 °C and 1 bar respectively. If the compression ratio is 18:1 and the maximum cycle temperature is 1700 k, determine, a) The pressure and temperature at the end of each process, b) The work done/kg/cycle, c) The thermal Efficiency.	L3	CO1	**10**
9.	a.	With a neat sketches explain the working principle of four stroke Petrol engine.	L2	CO2	**10**
		OR			
	b.	From the data given below, draw a heat balance sheet for a single cylinder four stroke oil engine, cylinder bore=200 mm, stroke= 350 mm, Gross mean effective pressure = 7 bar, pumping mean effective pressure=0.4 bar, engine speed= 5 rev, net brake load=370 N, brake drum radius=1.2 m, fuel used=3 kg/hr, Gross calorific value of oil=44000 kJ/kg, mass of cooling H_2O=4 kg/min, rise in temp of cooling=36 °C, mass of air supplied=1.4 kg/min, temp of exhaust gases= 320 °C, room temp=17 °C, mean specific heat of dry gasses=0.75, mass of exhaust gasses=2 kg/min.	L3	CO2	**10**
10.	a.	Derive the expression for work done in single stage compressor with clearance.	L3	CO3	**10**

		OR			
	b.	An air compressor takes in air at 1 bar ad 20 °C and compresses the same according to the law $PV^{1.2=}C$. It is then delivered to a receiver at constant pressure of 10 bar. If R for air = 0.287 kJ/kgK. Determine a) Temperature at the end of compression, b) Work done and heat transfer during compression per kg of air.	L3	CO3	**10**
11.	**a.**	Derive condition for maximum discharge through steam nozzle.	L3	CO4	**10**
		OR			
	b.	A nozzle is to be designed to expand steam at the rate of 0.10 kg/s from 500 kPa, 210 °C to 100 kPa. Neglect inlet velocity of steam. For a nozzle efficiency of 0.9, determine the exit area of the nozzle.	L3	CO4	**10**
12.	**a.**	With neat a sketch and p-h diagram explain the working of Li-Br vapor absorption refrigeration (VAR) cycle.	L2	CO5	**10**
		OR			
	b.	An air refrigerator was designed to produce 80 tons of refrigeration with air entering the compressor at 8 °C. The air was cooled after compression in a cooler to 27 °C. It was observed that the actual power required was 20% more than theoretical power with an air circulation rate of 2 kg/s. determine a) The theoretical COP, b) Actual COP, c) Power required to run compressor. Assume γ=1.4, Cp=1.005KJ/kg for air and the cycle is ideal.	L3	CO5	**10**
13.	**a.**	The following data refers to four stroke diesel engine cylinder diameter=200 mm, stroke=300 mm, speed=300 rpm, effective rake load=500 kg, mean circumference of the brake drum=400 mm, mean effective pressure=6 bar, diesel oil consumption=0.1 m^3/min, specific gravity of diesel=43900 KJ/kg. Find: a) BP, b) IP, c) FP, D) Mechanical Efficiency, e) Brake thermal efficiency and f) Indicated thermal efficiency.	L3	CO2	**10**
		OR			
	b.	The pressure and temperature at the beginning of compression in an air standard Otto cycle are 102 kpa and 315K. The heat is added during process at the rate of 250 KJ/Kg of air and air is used with a compression ratio of 9. Assuming γ=1.4 and R=287 J/kgK. Determine, a) Thermal efficiency of the cycle, b) Maximum cycle temperature and c) Maximum cycle pressure.	L3	CO1	**10**